Topic 1 — Independent Practice Part 1 Solutions

Brief checks. Use **We Need** → **We Know** → **We Solve** on your notebook.

Lesson 1-1: Relate Integers and Opposites

Question 1

We Need: The sum of an 18 m rise and an 18 m descent (Part A), and why the two changes are additive inverses (Part B).
We Know: Rising is positive and descending is negative. Additive inverses have the same size and opposite signs, so they add to zero.
We Solve: $+18 + (-18) = 0$
Answer: Part A: $+18 + (-18) = 0$; Part B: equal magnitude and opposite signs, so the two changes cancel

Question 2

We Need: An equation with $|a|$ and $|b|$.
We Know: $a + b = 0$ means $b = -a$, so the distances from zero match.
We Solve: $|a| = |b|$
Answer: $|a| = |b|$ (or equivalent)

Question 3

We Need: $|-9|$
We Know: Absolute value is distance from zero.
We Solve: $|-9| = 9$
Answer: 9

Question 4

We Need: Order -3, 8, -1 least to greatest.
We Know: On a number line, more left means smaller.
We Solve: -3, -1, 8
Answer: -3, -1, 8

Lesson 1-2: Understand Rational Numbers

Question 1

We Need: Decimal for $\frac{3}{8}$.
We Know: Divide $3 \div 8$.
We Solve: $3 \div 8 = 0.375$
Answer: 0.375

Question 2

We Need: Which outcome is *not* possible when $\frac{x}{y}$ (with $y \neq 0$) is converted by long division.
We Know: A fraction with a nonzero denominator always gives a decimal that either terminates or repeats.
We Solve: C is the impossible one: a decimal that neither terminates nor repeats is irrational, so it cannot come from a fraction.
Answer: C

Question 3

We Need: Who is correct about $0.\overline{27}$, Jordan or Casey.
We Know: Let $x = 0.\overline{27}$. Then $100x - x = 99x$, and the repeating tails cancel.
We Solve: $100x = 27.\overline{27}$, so $99x = 27$ and $x = \frac{27}{99} = \frac{3}{11}$. A repeating decimal is rational.
Answer: Jordan; $0.\overline{27} = \frac{3}{11}$

Lesson SC-1: Compare Rational Numbers

Question 1

We Need: Two comparisons of 10.4 and 10.6.
We Know: On a number line, more right means greater.
We Solve: $10.4 < 10.6$ and $10.6 > 10.4$
Answer: $10.4 < 10.6$ and $10.6 > 10.4$ (pounds OK)

Question 2

We Need: Symbols that fit $\frac{3}{4}$ and 0.75.
We Know: $\frac{3}{4} = 0.75$
We Solve: $\frac{3}{4} = 0.75$ exactly, so of the four symbols offered ($<$, $>$, $=$, $\neq$) only $=$ is true.
Answer: $=$ (only)

Lesson 1-3: Add Integers

Question 1

Part A: Start at 0. Move right 10, left 4, left 8.
Part B: $10 + (-4) + (-8) = -2$
Answer: Part A: number-line moves as above; Part B: -2

Question 2

We Need: The sum of the three signed moves.
We Know: Combine the positives and negatives.
We Solve: $15 + (-20) + 3 = -2$
Answer: -2

Question 3

We Need: Balance after 6 months.
We Know: Fee \$12.50 each month; $6 \times \$12.50 = \$75$.
We Solve: $\$500 - \$75 = \$425$
Answer: \$425

Question 4

We Need: The sum.
We Know: Opposite signs: subtract the distances, keep the sign of the larger.
We Solve: $-12 + 19 = 7$
Answer: 7

Lesson 1-4: Subtract Integers

Question 1

We Need: Distance from $5°F$ to $-8°F$.
We Know: Distance $= |5 - (-8)| = 13$.
We Solve: Valid expressions include $|5 - (-8)|$, $|-8 - 5|$, $5 - (-8)$, $|-8| + 5$.
Answer: Distance $13°F$; any correct distance expressions

Question 2

We Need: Distance between the two heights.
We Know: Distance is the absolute value of the difference.
We Solve: $|150 - 120| = 30$. Valid: $|150 - 120|$, $|120 - 150|$, $150 - 120$.
Answer: Distance 30 ft; any correct distance expressions

Question 3

We Need: Distance between the two depths.
We Know: Distance is the absolute value of the difference.
We Solve: $|-25 - (-40)| = 15$. Valid: $|-25 - (-40)|$, $|-40 - (-25)|$, $40 - 25$.
Answer: Distance 15 ft; any correct distance expressions

Question 4

We Need: Total distance the elevator travels.
We Know: Distance between two floors is the absolute value of their difference; add the legs.
We Solve: $|8 - (-3)| = 11$ and $|-1 - 8| = 9$; $11 + 9 = 20$.
Answer: 20 floors

Lesson 1-5: Add and Subtract Rational Numbers

Question 1

We Need: Balance after 4 months.
We Know: Start \$200 (below \$250), so the fee applies each month.
We Solve: $200 - 15 = 185$; $185 - 15 = 170$; $170 - 15 = 155$; $155 - 15 = 140$.
Answer: \$140

Question 2

We Need: Temperature after the three drops.
We Know: Each drop is a negative change; add them to the start.
We Solve: $-3.75 - 2.4 - 1.85 = -8.00$
Answer: $-8°C$

Question 3

We Need: Distance between the two marks.
We Know: Distance is the absolute value of the difference; write the mixed number as a decimal first.
We Solve: $2\frac{1}{2} = 2.5$; $|2.5 - (-1.75)| = 4.25$.
Answer: 4.25 miles

Lesson 1-6: Multiply Integers

Question 1

Part A: Descent is negative: -15 feet per minute.
Part B: $(-15) \times 12 = -180$ feet (180 feet below sea level).
Answer: Part A: -15; Part B: -180 feet (below sea level)

Question 2

We Need: Depth after 4.5 minutes.
We Know: A descent rate is negative, so the product is negative.
We Solve: $(-250.50) \times 4.5 = -1127.25$ m $= -1.12725$ km ≈ -1 km.
Answer: -1 kilometer (nearest km; descent)

Question 3

We Need: $(-5)^2$
We Know: The parentheses mean the negative is part of the base.
We Solve: $(-5)^2 = (-5) \times (-5) = 25$
Answer: 25

Question 4

We Need: -5^2
We Know: Without parentheses, the exponent applies to 5 only.
We Solve: $-5^2 = -(5^2) = -25$
Answer: -25
Difference: parentheses square the negative; without them, square first, then make negative.

Lesson 1-7: Multiply Rational Numbers

Question 1

We Need: The product.
We Know: Do the parentheses first, then multiply.
We Solve: $-0.5(10 + 6) = -0.5 \times 16 = -8$
Answer: -8

Lesson 1-8: Divide Integers

Question 1

Part A: Unit rate $= (-168) \div 7 = -24$ feet per minute.
Part B: $(-24) \times 4 = -96$ feet (96 feet below sea level).
Answer: Part A: -24; Part B: -96 feet (below sea level)

Question 2

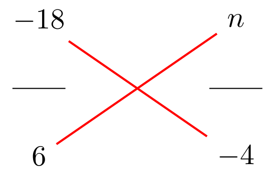
We Need: The quotient.
We Know: Opposite signs $\Rightarrow$ negative quotient.
We Solve: $(-96) \div 8 = -12$
Answer: -12

Question 3

We Need: The quotient.
We Know: Same signs $\Rightarrow$ positive quotient.
We Solve: $(-72) \div (-9) = 8$
Answer: 8

Question 4

We Need: n in $\dfrac{-18}{6} = \dfrac{n}{-4}$.
We Know: The fractions are equal. Use the butterfly method.



Butterfly: multiply along each red line.

We Solve: Cross products: $(-18)(-4) = 6 \cdot n$, so $72 = 6n$ and $n = 12$.
Check: $(-18) \div 6 = -3$ and $12 \div (-4) = -3$.
Answer: $n = 12$

Lesson 1-9: Divide Rational Numbers

Question 1

We Need: Each owner's share of the loss.
We Know: Split the loss equally among the 3 owners.
We Solve: $(-450.75) \div 3 = -150.25$
Answer: -150.25 (or $-\$150.25$ per owner)

Question 2

We Need: How many water stations.
We Know: $15\frac{1}{2} = \frac{31}{2}$, $3\frac{7}{8} = \frac{31}{8}$.
We Solve: $\frac{31}{2} \div \frac{31}{8} = \frac{31}{2} \times \frac{8}{31} = 4$ intervals. Stations at the end of each interval (including the finish line) $\Rightarrow$ 4.
Answer: 4 water stations

Lesson 1-10: Solve Problems with Rational Numbers

Question 1

We Need: The difference in elevation.
We Know: Subtracting a negative adds its opposite.
We Solve: $450 - (-120) = 450 + 120 = 570$
Answer: 570 feet

Lesson SC-2: Use the Laws of Exponents

Question 1

We Solve: $3^2 \times 3^4 = 3^{2+4} = 3^6 = 729$
Answer: 729 square inches

Question 2

Part A: $\dfrac{2^4}{2^2}$ or 2^{4-2} or 2^2
Part B: $16 \div 4 = 4$
Answer: Part A: $\dfrac{2^4}{2^2}$ (or equivalent); Part B: 4 cupcakes per friend

Question 3
We Solve: $(4^3)^2 = 4^{3\cdot 2} = 4^6 = 4096$
Answer: 4096

Question 4
We Solve: $7^0 = 1$
Answer: 1

Question 5
We Solve: $\dfrac{4^5}{4^2} \cdot 4^0 \cdot 4^{-1} = 4^{5-2} \cdot 1 \cdot \frac{1}{4} = 4^3 \cdot \frac{1}{4} = 64 \div 4 = 16$
Answer: 16

Question 6
We Solve: $(2^3)^2 = 2^6 = 64$, $(3 \cdot 2)^2 = 6^2 = 36$, and $\left(\frac{12}{2}\right)^2 = 6^2 = 36$. Then $64 \cdot 36 \div 36 = 64$.
Answer: 64

Question 7
We Solve: $5^0 = 1$, $3^{-2} = \frac{1}{9}$, and $\dfrac{2^6 \cdot 2^3}{2^7} = 2^{6+3-7} = 2^2 = 4$. Thus $1 + \frac{1}{9} + 4 = 5\frac{1}{9}$.
Answer: $\frac{46}{9}$ or $5\frac{1}{9}$

Question 8
We Solve: $\dfrac{7^4}{7^2} \cdot 7^0 = 7^{4-2} \cdot 1 = 7^2 = 49$
Answer: 49

Question 9
We Solve: $(3^2)^3 = 3^{2\cdot 3} = 3^6 = 729$
Answer: 729

Question 10
We Solve: $(2 \cdot 5)^3 = 10^3 = 1000$
Answer: 1000

Question 11
We Solve: $6^{-2} = \frac{1}{36}$ and $\left(\frac{8}{4}\right)^3 = 2^3 = 8$. Then $\frac{1}{36} + 8 = \frac{1}{36} + \frac{288}{36} = \frac{289}{36} = 8\frac{1}{36}$.
Answer: $\frac{289}{36}$ or $8\frac{1}{36}$

Question 12
We Solve: $3^4 \cdot 3^2 = 3^{4+2} = 3^6 = 729$
Answer: 729